Diwakar Mishra

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SUMMARY

B.Tech CSE-AI student (CGPA 9.43, top of branch) who builds and ships live full-stack applications - recommendation engines, graph algorithms, custom scoring pipelines, and LLM integrations, all deployed and publicly available.

EDUCATION

Maharaja Agrasen Institute of Technology 2024 – 2028
B.Tech in Computer Science & Engineering (Artificial Intelligence) CGPA: 9.43/10.0
PD Model School Class 12 – 94%
St. Karen's School Class 10 – 96.2%

SKILLS

- **Frontend:** React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Canvas API, D3/d3-force, Three.js
- **Backend:** Node.js, Express.js, Supabase (PostgreSQL), MongoDB, REST API Design, JWT, OAuth 2.0
- **Tools & Testing:** Git, GitHub, GitHub Actions (CI), Jest, VS Code, Vercel, Netlify
- **AI Engineering:** LLM Integration, Prompt Engineering, AI-Augmented Development
- **Programming:** C++, C, SQL

PROJECTS

Starweave - Interactive Influence Graph for Independent Music (Live Demo) | (GitHub)
Tech Stack: Next.js, TypeScript, react-force-graph, d3-force, Canvas API, Vercel

- Designed a citation-integrity data model over 297 artists and 1,053 influence edges - 908 cite a named source, 562 of them the artist's own words, plus 30 recorded artist denials
- Built a Dijkstra search weighted by citation quality, cutting poorly-sourced links on the average path from 70% to 11% while keeping paths nearly as short
- Laid out the genre lineage map with heavy-path decomposition and greedy interval-graph colouring, extending each interval by its label width so packed rows never collide

Arthouse Atlas - Curated Discovery Platform for World Cinema (Live Demo) | (GitHub)
Tech Stack: React.js, Node.js, Express.js, MongoDB, Tailwind CSS, JWT, Google OAuth 2.0, TMDB API

- Shipped a full-stack discovery platform for 497 curated arthouse films organised across directors, studios, and 12 cinematic movements, with region-filtered streaming availability via TMDB
- Built a recommendation engine scoring 200 candidate films per request against a per-user taste profile weighted by interaction type (favorites 3x, watched 1.5x, watchlist 1x) across mood tags, directors, movements, and decades
- Engineered a scoring pipeline rating every film on how arthouse and how canonical it is - from popularity inversion, auteur status, and weighted critical consensus - computed once at data-load and stored for fast filtering

CareerSaarthi - AI-Driven Career Accelerator & Mock Interview Platform (Live Demo) | (GitHub)
Tech Stack: React.js, Node.js, Supabase (PostgreSQL), Web Speech API, OpenRouter (GPT-4o-mini), Gemini API

- Engineered a real-time voice interview coach - spoken questions, live microphone transcription via Web Speech API, timed auto-submit, and automated AI scoring (clarity, STAR structure, keyword density) saved to Supabase
- Built resilient LLM integrations across 6 career tools: PDF resume parsing with GPT-4o-mini for ATS scoring, sessionStorage caching (30-min TTL) to absorb rate limits, and fallbacks preventing blank states on the Adzuna job board

ACHIEVEMENTS

- **LeetCode Contest Rating 1800+:** 350+ problems solved across arrays, graphs, trees, and dynamic programming. (view profile)
- **Flipkart GRiD 8.0 – Semi-Finalist:** Advanced through the coding and case-study rounds of Flipkart's national engineering competition.
- **CodeX Hackathon Winner:** Won the Neighbourly Choice Award at MAIT. (view certificate)